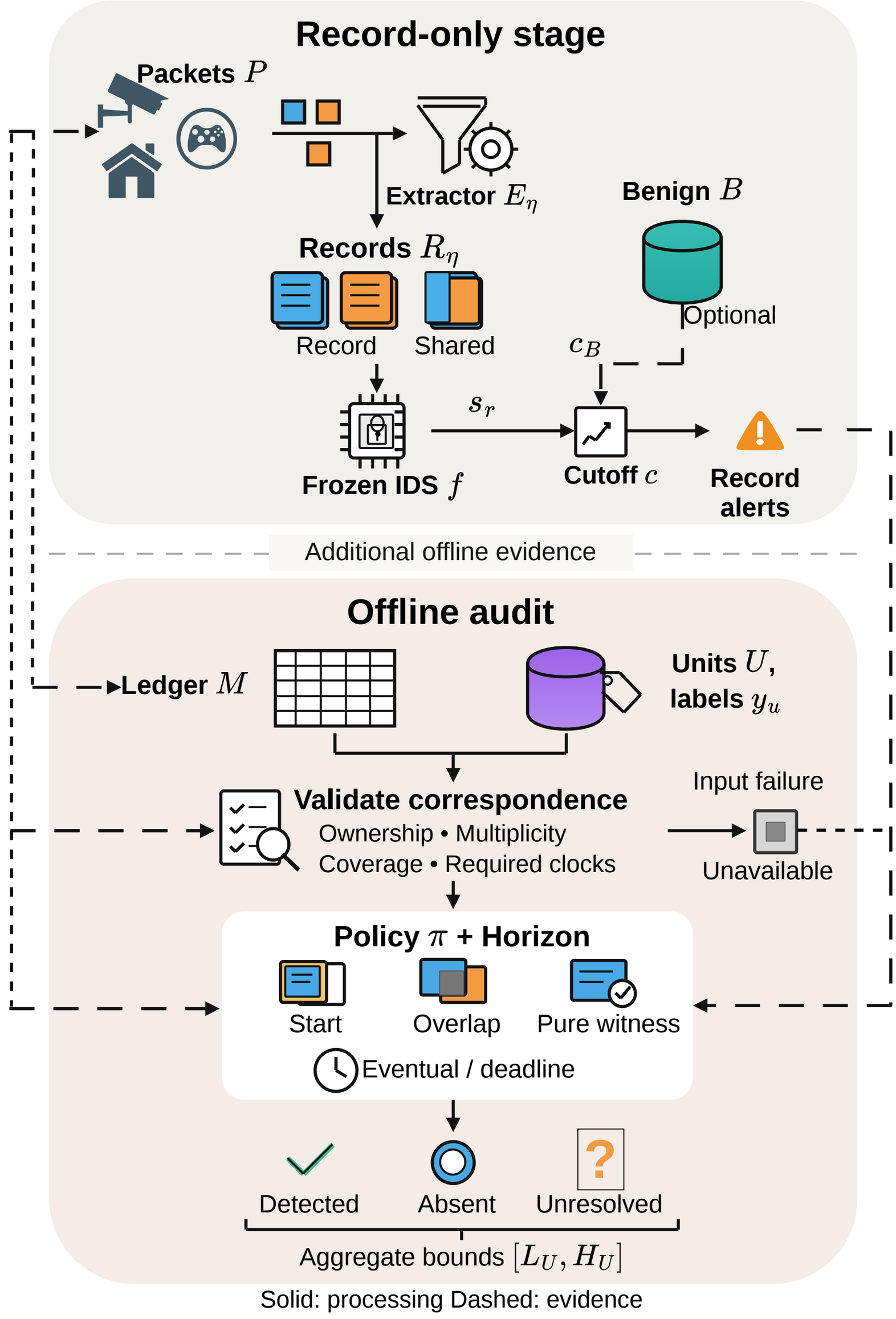

Record-only stage
Packets P
Extractor E_η
Records R_η
Record
Shared
Benign B
Optional
c_B
s_r
Frozen IDS f
Cutoff c
Record alerts
Additional offline evidence
Offline audit
Ledger M
Units U, labels y_u
Validate correspondence
Ownership • Multiplicity
Coverage • Required clocks
Input failure
Unavailable
Policy π + Horizon
Start
Overlap
Pure witness
Eventual / deadline
Detected
Absent
Unresolved
Aggregate bounds $[L_U, H_U]$
Solid: processing Dashed: evidence